

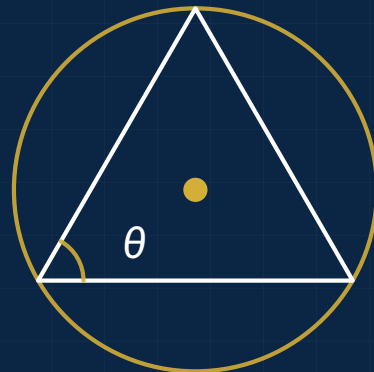


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CHAPTER 4 • GEOMETRY & MEASURES • TOPIC 2

Angles, Polygons & Parallel Lines

Every angle question is a chain of facts.
Write the angle, then the reason — on a
line, at a point, alternate, corresponding,
co-interior, or the polygon rules.



4

VARIANTS

16

PAST-PAPER PARTS

49

PRACTICE QUESTIONS

YOU WILL MASTER

V1 Angle facts & parallel lines

V2 Angle sum of a polygon

V3 Regular polygons: find n

V4 Polygons joined together

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Angles at a glance



Your notes

Angles, Polygons & Parallel Lines

LINES & TRIANGLES

V1 Angle facts & parallels

alternate, corresponding

PAST PAPERS: 3 OF 16

ONE POLYGON

V2 Angle sum

$$(n - 2) \times 180^\circ$$

PAST PAPERS: 2 OF 16

V3 Regular: find n

$$n = \frac{360}{\text{ext}}$$

PAST PAPERS: 4 OF 16

POLYGONS TOGETHER

V4 Polygons joined

$$360 - 108 - 96$$

PAST PAPERS: 7 OF 16

THE KEY HABIT**ANGLE, THEN REASON**

“Give a reason for each stage” — the reasons carry marks.

★ How often does each variant come up?

16 past-paper questions (Jan 2020 → Nov 2025); 49 practice parts — grouped by type.

	PAST PAPERS (OUT OF 16)	PRACTICE PARTS
V1 Angle facts, parallel lines	3	23
V2 Angle sum of a polygon	2	6
V3 Regular polygons: find n	4	7
V4 Polygons joined together	7	13

WHAT THIS TELLS YOU One angle question on almost every paper (Q3–Q15, 3–5 marks), most often two regular polygons sharing a side. Know the interior angles of the pentagon (108°), hexagon (120°) and octagon (135°) by heart.

BEFORE YOU START

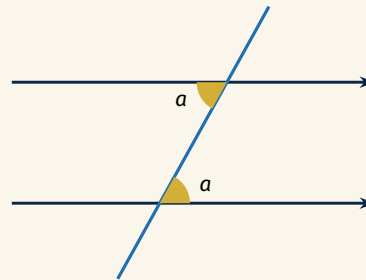
The big idea

**Angles are found in a chain.**

Each step uses one fact. Write the angle you found and the **reason** next to it — then the next angle follows.

Parallel lines give three pairs:

alternate (Z, equal), **corresponding** (F, equal), **co-interior** (C, add to 180°).



ALTERNATE (Z): EQUAL

The facts (with the words for reasons)
On a line 180°

“angles on a straight line”

At a point 360°

“angles around a point”

Triangle 180°

“angles in a triangle”

Quadrilateral 360°

“angles in a quadrilateral”

Polygon rules
Angle sum $(n - 2) \times 180^\circ$

Any n -sided polygon.

Exterior anglesadd to 360°

Regular: each $\frac{360}{n}$.

Interior + exterior 180°

Pentagon 108° , hexagon 120° , octagon 135° .

**CALCULATOR FIRST, THEN CHECK**

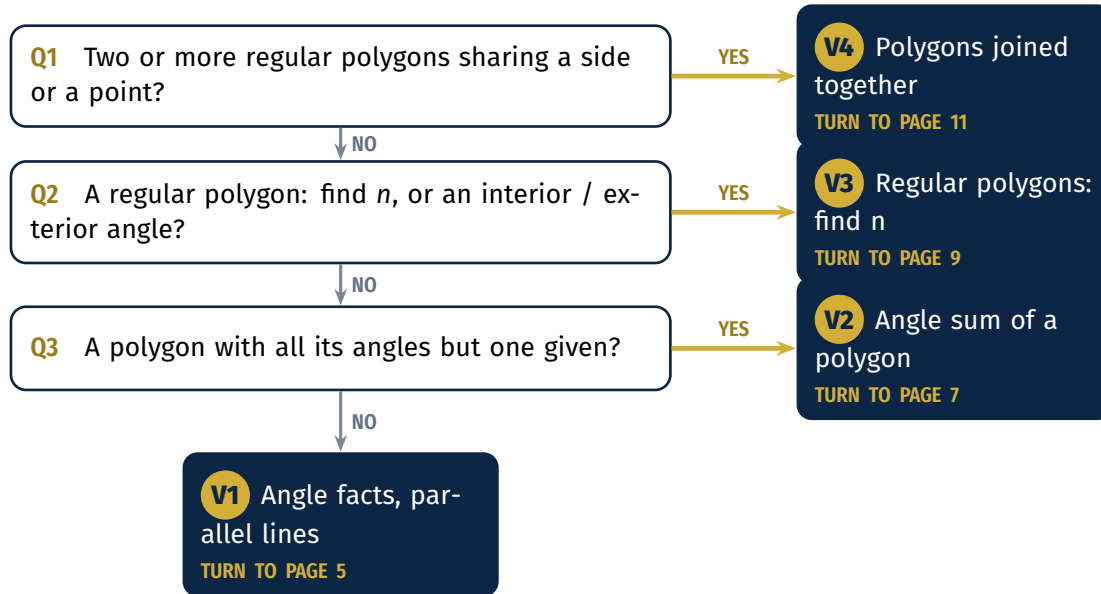
Check a regular polygon: interior + exterior must be 180° , and $n \times$ exterior must be 360° .

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1

Angle Facts & Parallel Lines

Find one angle at a time; name the fact each time (Z, F, C, isosceles, line, point).

SEEN IN PAST PAPERS

3 / 16

PRACTICE QUESTIONS: 23

 Your notes

Came up in: Jan 2020 P2H Q4 • May 2021 P2H Q5 • Jun 2023 P2H Q3

? What is it?

In plain words

Angles written with x : use a fact to make an **equation** (triangle: sum 180° ; parallel sides: co-interior add to 180°), solve, then work out each angle.

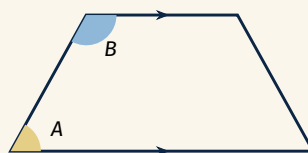
Parallel-line questions want the **reason** for each step.

IT LOOKS LIKE

$$(4x + 10)^\circ$$

trapezium

$$BE = CE$$



$$A + B = 180^\circ \text{ (co-interior)}$$

How do I do it?

1 Spot

Parallel arrows? Equal sides? A triangle?

2 Fact

Choose the fact that links the angles.

3 Equation

Solve for x if needed.

4 Answer + reason

Each angle with its reason; answer the angle asked.

ANGLE, THEN REASON Z equal, F equal, C add to 180.

V1 Angle Facts & Parallel Lines

VARIANT 1 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

Parallel sides with angles in x → **Co-interior: sum = 180°**

" $BE = CE$ " → **Isosceles: base angles equal**

"Give a reason for each stage" → **Write the fact after every angle**

**WORKED EXAMPLE** **PAST PAPER**

June 2023 · Paper 2H · Q3 · 4 marks

$ABCD$ is a trapezium with BC parallel to AD . Angle $A = (4x - 27)^\circ$, angle $B = (3x + 46)^\circ$, angle $D = (3x + 10)^\circ$. Find the size of the largest angle inside the trapezium.

STEP 1 $(4x - 27) + (3x + 46) = 180$ ◀ co-interior angles

STEP 2 $7x + 19 = 180 \Rightarrow x = 23$ ◀ solve

STEP 3 $A = 65$, $B = 115$, $D = 79$, $C = 360 - 259 = 101$ ◀ each angle; quadrilateral 360

STEP 4 largest = 115 ◀ compare

ANSWER 115° ✓ $C + D = 180$ too ✓**WATCH OUT**

✗ $A = B$ (alternate)

✓ $A + B = 180$

Between the parallel sides they are co-interior (a C shape), not a Z.

**EXAM TIP**

Find all four angles — the question asks for the largest, not for x .

**Check yourself**

calculator first · answers on page 13

1.1

Jan 2020 P2H

A triangle has angles 30° , $(4x + 10)^\circ$ and $(x + 20)^\circ$. Find x .

Answer:

1.2

May 2021 P2H

$BC \parallel GH$, $BE = CE$, angle $BEC = 44^\circ$; $EBGJ$ is straight. Find angle JGH (exterior at G).

Answer:

1.3

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Parallel lines cut by a transversal; one angle is 64° . Find its co-interior angle.

Answer:

V2

VARIANT 2

Angle Sum of a Polygon

Sum = $(n - 2) \times 180^\circ$; subtract the angles you know.

SEEN IN PAST PAPERS

2 / 16

PRACTICE QUESTIONS: 6



Your notes

Came up in: Oct/Nov 2020 P1H Q8 • Jun 2022 P1H Q23

? What is it?

In plain words

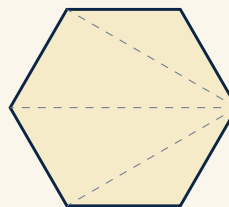
Any polygon splits into $n - 2$ **triangles** from one corner, so its angles add to $(n - 2) \times 180^\circ$.

Missing angle: total minus the ones you know. Unknown n : set the sum equal to $(n - 2)180$ and solve.

IT LOOKS LIKE

10-sided, one angle x

angles in an AP

6 sides $\rightarrow$ 4 triangles $\rightarrow 720^\circ$

How do I do it?

1 CountNumber of sides n .**2 Total** $(n - 2) \times 180$.**3 Known**

Add the angles you have.

4 Missing

Total - known.

 $(n - 2) \times 180$ Pentagon 540° , hexagon 720° .

V2 Angle Sum of a Polygon

VARIANT 2 · SEE IT DONE, THEN TRY

 Your notes
 WHEN YOU SEE → YOU DO

"Here is a 10-sided polygon ... work out x " → $8 \times 180 = 1440$, **subtract**

"the interior angles ... are the terms of an arithmetic sequence" → $S_n = (n - 2)180$: **solve for n**

"Zak says ... is he correct?" → **Compare with $(n - 2)180$**

 WORKED EXAMPLE **PAST PAPER**

Oct/Nov 2020 · Paper 1H · Q8 · 4 marks

A 10-sided polygon has angles 148° , 150° , 150° , 168° , 168° , 134° , 134° , 125° , 125° and x . Work out the value of x .

STEP 1 $(10 - 2) \times 180 = 1440$ ◀ angle sum

STEP 2 $148 + 2(150) + 2(168) + 2(134) + 2(125)$ ◀ the known angles

STEP 3 $= 1302$ ◀ add

STEP 4 $x = 1440 - 1302 = 138$ ◀ missing angle

ANSWER $x = 138$ ✓ $1302 + 138 = 1440$ ✓ **WATCH OUT** $\times 10 \times 180 = 1800$ ✓ $(10 - 2) \times 180 = 1440$ It is $n - 2$ triangles, not n . **EXAM TIP**

Pair up the equal angles to add quickly and safely.

 Check yourself

calculator first · answers on page 13

2.1

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Work out the sum of the interior angles of a 9-sided polygon.

Answer:

2.2

Jun 2022 P1H

The interior angles of an n -sided polygon are 177° , 175° , 173° , ... (an arithmetic sequence). Find n .

Answer:

2.3

Practice · Very hard

A pentagon has three angles of 144° and two equal angles w . Find w .

Answer:

V3

VARIANT 3

**Regular Polygons:
Find n**

$$\text{Exterior} = 180 - \text{interior}; n = \frac{360}{\text{exterior}}.$$

SEEN IN PAST PAPERS

4 / 16

PRACTICE QUESTIONS: 7



Your notes

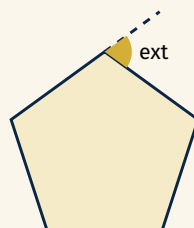
Came up in: Jun 2022 P1H Q4 • May 2023 P1HR Q15 • Jun 2025 P1H Q3 • Nov 2025 P1H Q12

? What is it?**In plain words**

In a regular polygon all the **exterior** angles are equal and add to **360°** .

So exterior = $\frac{360}{n}$ and $n = \frac{360}{\text{exterior}}$.

Given the interior angle? Exterior = $180 - \text{interior}$ first.

IT LOOKS LIKEinterior 162° $(9x + 27)^\circ$ and $(2x - 23)^\circ$ 

$$\text{ext} = \frac{360}{5} = 72^\circ, \text{ int} = 108^\circ$$

How do I do it?**1 Exterior**

180– interior (or from the diagram).

2 Divide

$$n = 360 \div \text{exterior}.$$

3 Whole? n must be a whole number ≥ 3 .**4 Check**

$$n \times \text{exterior} = 360.$$

 $n = 360 \div \text{EXTERIOR}$ Not $360 \div \text{interior}$.

V3 Regular Polygons: Find n

VARIANT 3 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"Each interior angle is 162° . How many sides?" → **Exterior 18° , $n = 20$**

Interior $(9x + 27)^\circ$, exterior $(2x - 23)^\circ$ → **Add to 180: find x**

Three regular polygons meet at a point → **Their angles add to 360°**

WORKED EXAMPLE **PAST PAPER**

November 2025 · Paper 1H · Q12 · 4 marks

Part of a regular n -sided polygon has an interior angle of $(9x + 27)^\circ$ and an exterior angle of $(2x - 23)^\circ$. Work out the value of n .

STEP 1 $(9x + 27) + (2x - 23) = 180$ ◀ interior + exterior

STEP 2 $11x + 4 = 180 \Rightarrow x = 16$ ◀ solve

STEP 3 exterior = $2(16) - 23 = 9^\circ$ ◀ substitute

STEP 4 $n = 360 \div 9 = 40$ ◀ divide

ANSWER $n = 40$ ✓ interior 171° , $171 + 9 = 180$ ✓**WATCH OUT**✗ $n = 360 \div 171$ ✓ $n = 360 \div 9$

Divide 360 by the exterior angle.

EXAM TIP

If n is not a whole number, recheck x — a regular polygon can't have 7.3 sides.

Check yourself

calculator first · answers on page 13

3.1

Jun 2022 P1H

Regular polygons A, B, C meet at a point with angles $8x^\circ$, $7x^\circ$, $3x^\circ$. B has n sides ($7x^\circ$ is its interior angle). Find n .

Answer:

3.2

Practice · Medium

Each interior angle of a regular polygon is 156° . How many sides?

Answer:

3.3

Practice · Medium

Find the exterior and interior angles of a regular 12-sided polygon.

Answer:

V4

VARIANT 4 MOST COMMON IN THE EXAM

Polygons Joined Together

Interior angles of each regular polygon, then angles at a point, on a line, or in an isosceles triangle.

SEEN IN PAST PAPERS

7 / 16

PRACTICE QUESTIONS: 13

 Your notes

Came up in: Jan 2021 P1HR Q8 • Nov 2021 P2H Q5 • Jan 2022 P2HR Q4 • Jan 2022 P2H Q11 • Jun 2023 P2H Q11 • May 2024 P1H Q5 • Nov 2024 P2H Q11

? What is it?

In plain words

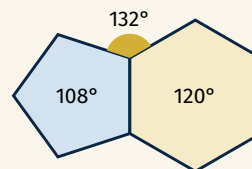
Work out each polygon's **interior angle** first and write it on the diagram.

Then use **angles at a point** (360°), **on a line** (180°), or the **isosceles triangle** made when two polygons share a side (equal sides!).

IT LOOKS LIKE

pentagon + hexagon

octagon + pentagon



$$\text{gap} = 360 - 108 - 120 = 132^\circ$$

How do I do it?

1 Interiors

Pentagon 108, hexagon 120, octagon 135, decagon 144.

2 Label

Write them on the diagram.

3 Link

At a point 360; on a line 180; difference of two angles.

4 Isosceles?

Shared side length $\rightarrow$ equal base angles: $\frac{180 - \text{apex}}{2}$.

INTERIORS FIRST The most common variant here.

V4 Polygons Joined Together

VARIANT 4 · SEE IT DONE, THEN TRY

 Your notes

WHEN YOU SEE → YOU DO

"ABCDE is a regular pentagon ... angle AEF = 96°" → **At E: $360 - 108 - 96$**

"regular octagon and regular pentagon" sharing a side → **$135 - 108 = 27^\circ$; isosceles**

"AF is a straight line" → **Triangle with two equal sides**

WORKED EXAMPLE PAST PAPER

May 2024 · Paper 1H · Q5 · 4 marks

ABCDE is a regular pentagon. Angle AEF = 96°. Work out the size of the obtuse angle FED.

STEP 1 interior = $180 - \frac{360}{5} = 108^\circ$ ◀ regular pentagon

STEP 2 AED = 108° ◀ one interior angle

STEP 3 FED = $360 - 108 - 96$ ◀ angles around a point E

STEP 4 = 156° ◀

ANSWER 156°

✓ $108 + 96 + 156 = 360$ ✓

WATCH OUT

✗ $180 - 108 - 96$

✓ $360 - 108 - 96$

The three angles are around a point, not on a line.

EXAM TIP

Write the reason ("angles around a point") — it is part of the method mark.

Check yourself

calculator first · answers on page 13

4.1 Jan 2022 P2HR

A regular octagon and a regular pentagon share side AB. In isosceles triangle IBC the apex angle is $135 - 108$. Find x (a base angle).

Answer:

4.2 Nov 2024 P2H

A regular pentagon ABCDE and a regular hexagon DEFGHI share DE. Find angle EAF.

Answer:

4.3 Jan 2021 P1HR

Regular pentagon ABCDE and regular hexagon CFGHID share CD; AEJ and HIJ are straight. Find angle x at J.

Answer:

CHECK YOURSELF

Answers



Your notes

Every answer below was checked with sympy (check_answers.py).

1.1 $x = 24$

1.2 112°

1.3 116°

2.1 1260°

2.2 $n = 18$

2.3 $w = 54^\circ$

3.1 $n = 9$

3.2 15 sides

3.3 $30^\circ, 150^\circ$

4.1 76.5°

4.2 24°

4.3 96°

THE WHOLE TOPIC ON ONE PAGE

Angles cheat sheet



Your notes
V1 Facts, parallels 3/16 $Z, F \angle; C + \angle = 180$

Angle, then reason.

V2 Angle sum 2/16 $(n - 2) \times 180$

Total minus the known.

V3 Find n 4/16

$$n = \frac{360}{\text{ext}}$$

Exterior = $180 - \text{interior}$.**V4 Joined 7/16**

$$360 - 108 - 96$$

Interiors first; point, line, isosceles.

**TOP 3 LOST MARKS**

1. no reasons written ×
2. $360 \div \text{interior}$ ×
3. $n \times 180$ for the angle sum ×

BOTTOM LINE Angle, then reason · $(n - 2) \times 180$ · exterior = $\frac{360}{n}$ · 108, 120, 135, 144.